



Interfacial electrochemical investigation of 3D space-confined MnFe_2O_4 for high-performance ionic liquid-based supercapacitors

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ABSTRACT

Although MnFe_2O_4 is regarded as superior supercapacitors (SCs) electrodes in aqueous electrolytes owing to its diatomic synergistic effect and electron-occupied states at Fermi level, researches on MnFe_2O_4 electrodes for ionic liquid (IL)-based SCs which can easily achieve high energy densities, are still hindered because of mediocre interfacial electrochemical behaviors of MnFe_2O_4 electrodes in IL electrolyte with high viscosity. Herein we propose a 3D space-confined MnFe_2O_4 electrode (denoted as 3DSC-MFQDs) and conduct in-depth research on its interfacial electrochemical behaviors in high-voltage IL electrolyte. The formation mechanism of the Stern/diffusion layer of IL ions near electrode surface has been studied for the first time. Investigation of potential-driven ion accumulation deeply elucidates strong characteristic adsorption between IL ions and electrode with abundant active sites which can easily overlap with the electron cloud of positively charged imidazole rings of EMIM⁺ cations. Stronger adsorption improves the interfacial electrochemical behavior of 3DSC-MFQDs (high diffusion coefficient of $3.374 \times 10^{-8} \text{ m}^2/\text{s}$), providing highly-reversible redox reaction of 3DSC-MFQDs and thus leading to durable structures of 3DSC-MFQDs electrode. On this basis, flexible SCs (FSCs) with high energy/power density based on ionogel electrolyte are fabricated and act as efficient flexible energy storage devices for practical applications.

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Author contribution

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1. Introduction

Benefiting from unique diatomic synergistic effect and efficient electron-occupied states at the Fermi level, bimetallic oxides exhibit favorable electrical conductivity, enhanced redox couples and accessible active sites, hence they have received considerable attention in the field of energy storage applications [1,2]. Among them, ferromanganese bimetallic oxide (MnFe_2O_4), a novel spinel ferrite, has been deemed to be a promising electrode for high-efficiency supercapacitors (SCs) due to its multiple valence, high electrochemical activity and fast valence-state response capability [3,4]. In recent years, efforts have been dedicated to the research of MnFe_2O_4 with various structures for SCs electrodes [4–7]. These as-obtained MnFe_2O_4 electrodes dominantly showed faster redox reaction and much higher pseudocapacitance than monometallic oxides (Mn_xO_y or Fe_xO_y). Although MnFe_2O_4 electrodes have been documented to be utilized in several high-performance SCs based

on aqueous electrolytes (KOH, KCl, Na₂SO₄ etc.), it is challenging to attain high energy density of these devices due to the narrow electrochemical windows of aqueous electrolytes, which is enforced by the low decomposition voltage of water around 1.23 V.

Ionic liquid (IL), a class of room-temperature molten salts, has been regarded as an attractive candidate to substitute aqueous electrolytes of SCs. This is owing to the unique properties of IL, such as wide operation voltage (>2.5 V), negligible vapor pressure, nonflammability, and high thermal stability [8–11]. Currently, extensive researches have been focusing on the development of SCs using IL electrolytes, especially carbon-based electrical double-layer SCs [12–14]. Compared with simple double-layer capacitance of carbon electrodes, Faradaic pseudocapacitance stemming from reversible redox reactions of pseudocapacitive electrodes can deliver much higher specific capacitance for SCs [15,16]. However, although MnFe₂O₄ electrodes have been demonstrated to possess large specific capacitance in aqueous electrolytes by virtue of redox reactions, so far the reports on MnFe₂O₄-IL electrolyte system are relatively limited till date. The main reason could be that the poor interfacial wettability and mass diffusion between MnFe₂O₄ electrodes and IL electrolyte with high viscosity hinder their application in high-voltage IL electrolyte [17].

A desirable approach is to confine the MnFe₂O₄ in a suitable nanospace to improve its interfacial electrochemical behaviors. In this work, we report a 3D space-confined MnFe₂O₄ quantum dots (denoted as 3DSC-MFQDs) electrode, which is realized by suitable confinement of MnFe₂O₄ quantum dots (MFQDs) in 3D nanospace. This 3D nanospace is supported by metal organic frameworks (MOFs) derived porous carbon polyhedrons (PCPs). In such an electrode, conductive PCPs act as ion-accessible channel and charge-transfer pathway, allowing high motion dynamics of IL ions and electrons. In addition, the active sites of the electrode can be greatly increased owing to the high porosity and heteroatoms doping of MOFs-derived PCPs [18–20]. In order to figure out the interfacial electrochemical behaviors of 3DSC-MFQDs, we went into the formation mechanism of the Stern layer and diffusion layer of IL ions in an electric field and specifically near the charged electrode surface for the first time. Further, by the measurement of differential capacitance, potential-driven ion accumulation probe elucidates that strong adsorption occurs between IL cations (EMIM⁺) and electrode surface with abundant active sites, especially N defects in PCPs, which can easily overlap with the electron cloud of the positively charged imidazole ring of the EMIM⁺ cation. Similar characteristic adsorption may appear between 3DSC-MFQDs electrode and other imidazole-based IL electrolytes. From this base, the stronger adsorption between EMIM⁺ cations and electrode surface improves the interfacial electrochemical behavior of 3DSC-MFQDs (a high diffusion coefficient of 3.374×10^{-8} m²/s), providing sufficient and highly reversible redox reaction of MFQDs and thus leading to the high specific capacitance of 3DSC-MFQDs electrode. Moreover, combining with ex-situ XRD analysis with electrochemical testing, it is indicated that the durable structural stability of 3DSC-MFQDs electrode has been achieved during the charging-discharging process.

As a proof of concept, high-performance flexible SCs (FSCs) based on ionogel electrolyte have been fabricated, representing a high energy density (67.6 W h/kg) at a high power density (1.75 kW/kg), as well as remarkable rate and cycling performance. For a real application, the FSCs can effectively power various wearable electronics and reveal their promising electrochemical reliability under hard mechanical conditions.

2. Experimental

2.1. Synthesis of 3DSC-MFQDs

All of the reagents were analytical grade without further purification. Firstly, PCPs were derived from a zeolitic imidazole metal-organic framework (ZIF-8) according to the previous literature [21]. After that, 100 mg PCPs were uniformly dispersed in 80 mL of ethanol under sonication for 2 h. Then, 1 mM Mn(NO₃)₂ and 2 mM Fe(NO₃)₃·9H₂O were dissolved separately in the above solution and stirred for 5 min. 30 wt% ammonia aqueous solution was used to adjust the solution to pH 9–10. Subsequently, the mixed suspension was transferred to a 100 mL Teflon-lined stainless-steel autoclave and heated at 180 °C for 24 h. Finally, the solid product was centrifuged, washed with deionized water, freeze-dried and then calcined at 300 °C in Ar atmosphere for 2 h to form 3DSC-MFQDs. For comparison, pure MnFe₂O₄ quantum dots (denoted as MFQDs) were also prepared through a procedure similar to that mentioned above without addition of PCPs.

2.2. Characterization methods

Field emission scanning electron microscope (FESEM, FEI Sirion 200) and Transmission electron microscopy (TEM, JEM-2010F) were carried out to characterize the morphology of the samples. X-ray diffraction (XRD) patterns were characterized on a powder XRD system with Cu K α radiation. X-ray photoelectron spectroscopic (XPS) measurements were performed on a Kratos AXIS Ultra DLD spectrometer with an Al K α X-ray source. Nitrogen absorption and desorption measurements were performed with an Autosorb IQ instrument at 77 K. The specific surface areas were calculated by the Brunauer-Emmett-Teller (BET) method, and the pore size distributions were determined from the adsorption branch of the isotherms based on the density functional theory (DFT).

2.3. Electrochemical measurements

All the electrochemical measurements were carried out using a VMP3 multi-functional electrochemical analysis instrument (Bio-Logic, France). Electrochemical behaviors were investigated by cyclic voltammetry (CV), galvanostatic charge-discharge (GCD) and electrochemical impedance (EIS) methods. The CV and GCD tests were measured at various scan rates and current densities. The EIS plots were performed in the frequency ranging from 0.01 Hz to 100 kHz with 5 mV AC amplitude. All tests were conducted using the three-electrode systematic method in 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF₄) IL electrolyte. The platinum (Pt) foil and Ag/AgCl electrodes were used as the counter and reference electrodes, respectively. The working electrode was prepared by mixing our samples (3DSC-MFQDs or MFQDs) with acetylene black and polyvinylidene fluoride (PVDF) as the binder dissolved in *N*-methyl-2-pyrrolidone (NMP) in a weight ratio of 8:1:1. The slurry was pasted on nickel foam substrate using a medical blade, which was then dried in a vacuum oven at 80 °C overnight to obtain electrode. Each electrode contains active material with a mass loading of about 1.2 mg/cm². For FSCs application, a two-electrode cell configuration was assembled using ionogel electrolyte, in which graphite paper functioned as flexible current collector, along with 3DSC-MFQDs and commercial activated carbon (AC) served as negative and positive electrodes, respectively. Then, the ionogel electrolyte was slowly sandwiched between two electrodes, which were pressed together under a pressure of ~1 MPa for 1 h. Ionogel electrolyte was prepared with 0.5 g Poly(vinylidene fluoride-hexafluoropropylene) (P(VDF-HFP)), 2 g EMIMBF₄ IL and 5 mL of acetone solution.

2.4. Calculations for the electrochemical tests

According to previous reports [22,23], the specific capacitance of a single electrode was evaluated using:

$$C = I \cdot \Delta t / s \cdot \Delta V \quad (1)$$

where C (mF/cm^2) is the specific capacitance, I (mA) is the discharge current, Δt (s) is the discharge time, ΔV (V) is the voltage variation during the discharge process after “IR drop”, s (cm^2) is the real area of the electrode.

The energy density of assembled FSCs was calculated on the basis of the discharge branch of GCD curve by using Equation (2) and the power density was evaluated by using Equation (3):

$$E = C_f \cdot (\Delta V)^2 / 7.2 \quad (2)$$

$$P = 3600 \cdot E / \Delta t \quad (3)$$

where C_f (F/g) is the specific capacitance of the FSC, E (Wh/kg) is the energy density, P (W/kg) is the power density, Δt (s) is the discharge time, ΔV (V) is the voltage window of the full FSC.

2.5. Ex-situ X-ray diffraction test

Before starting the ex-situ XRD test, CV scanning with a rate of 50 mV/s has been conducted for 50 cycles to stabilize the electrochemical signature of the electrodes. For ex-situ XRD experiment, electrodes were charged and discharged at a current density of 8 mA/cm^2 , and the stop positions of the GCD test are shown in Fig. 4h and Figs. S11. Once the electrodes were charged and discharged to a desired voltage point, the GCD test paused and the electrodes were lifted out of the electrolyte solution to be washed

clean and then tested by XRD analysis. After XRD analysis, the electrodes continued the unfinished GCD test until the next desired voltage point, and were lifted out of the IL electrolyte to be washed clean and then tested by XRD analysis. Above steps were repeated until getting all the XRD patterns of 3DSC-MFQDs and MFQDs electrodes at various constant potentials.

3. Results and discussion

Fig. 1a shows the size and morphology of the PCPs derived from ZIF-8. 3D rhombic dodecahedral morphology can be observed for PCPs with an average size of 500 nm, while the abundant and continuous porous structure exists in the PCPs (Figs. S1 and 2), providing numerous facile accessible channels and short diffusion distance for electrolyte ions [24,25]. Owing to the PCPs with 3D robust and porous structure, a large number of MFQDs uniformly grow in the compartment of PCPs to form 3DSC-MFQDs (Fig. 1b) [26]. The 3D nanospace supported by PCPs is favorable for the confinement of MFQDs (Fig. 1c), in which the size of MFQDs is about 5 nm (Fig. 1d). High-resolution TEM (HRTEM) image in Fig. 1e shows the lattice fringes with spacing of 0.256 nm corresponding to the (311) plane of MnFe_2O_4 in MFQDs [27], whereas the disordered area is ascribed to the PCPs, indicating that the MFQDs are totally confined in the PCPs backbone. Furthermore, XRD pattern (Fig. 1f) exhibits typical characteristic peaks indexed to iwakiite MnFe_2O_4 (JCPDS no. 38–0430) of MFQDs, which are consistent with the result of HRTEM.

As shown in Fig. 2a, the corresponding elemental mapping analysis of the constituting elements C, N, Mn, Fe and O clearly displays a well-defined compositional profile of 3DSC-MFQDs. XPS full spectrum (Fig. 2b) reveals that the 3DSC-MFQDs contains C, N, Fe, Mn and O atoms with a high N content of 5.2 wt%. As shown the high-resolution N 1s spectrum in Fig. 2c, it can be fitted into three subpeaks located at 398.8, 400.0 and 401.5 eV, which are ascribed

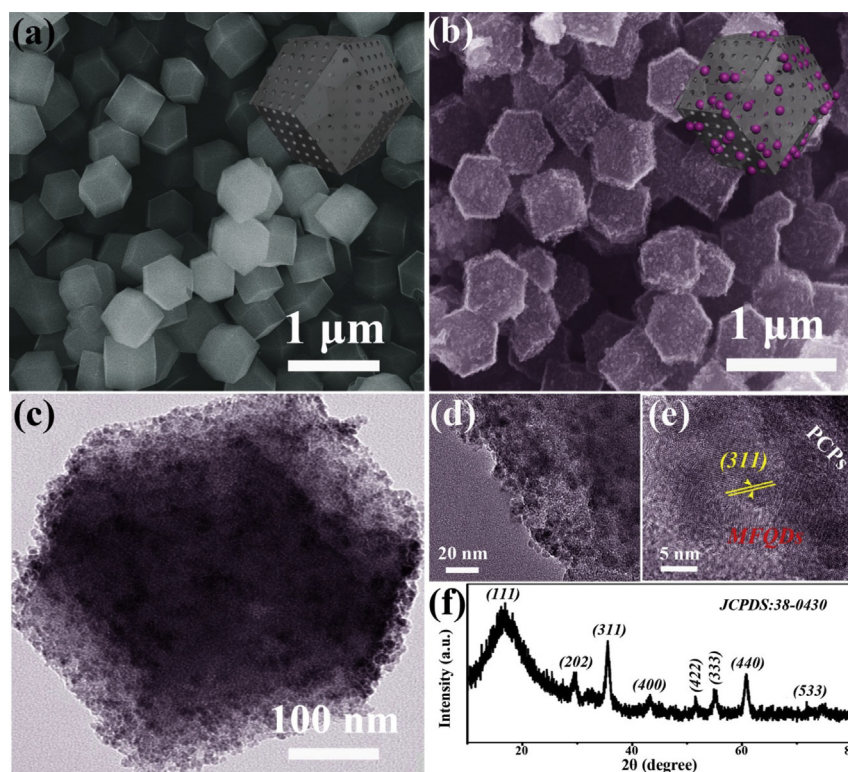


Fig. 1. (a) FESEM image of PCPs; (b) FESEM image; (c–d) TEM images with different magnification; (e) High-resolution TEM (HRTEM) image and (f) XRD pattern of 3DSC-MFQDs.

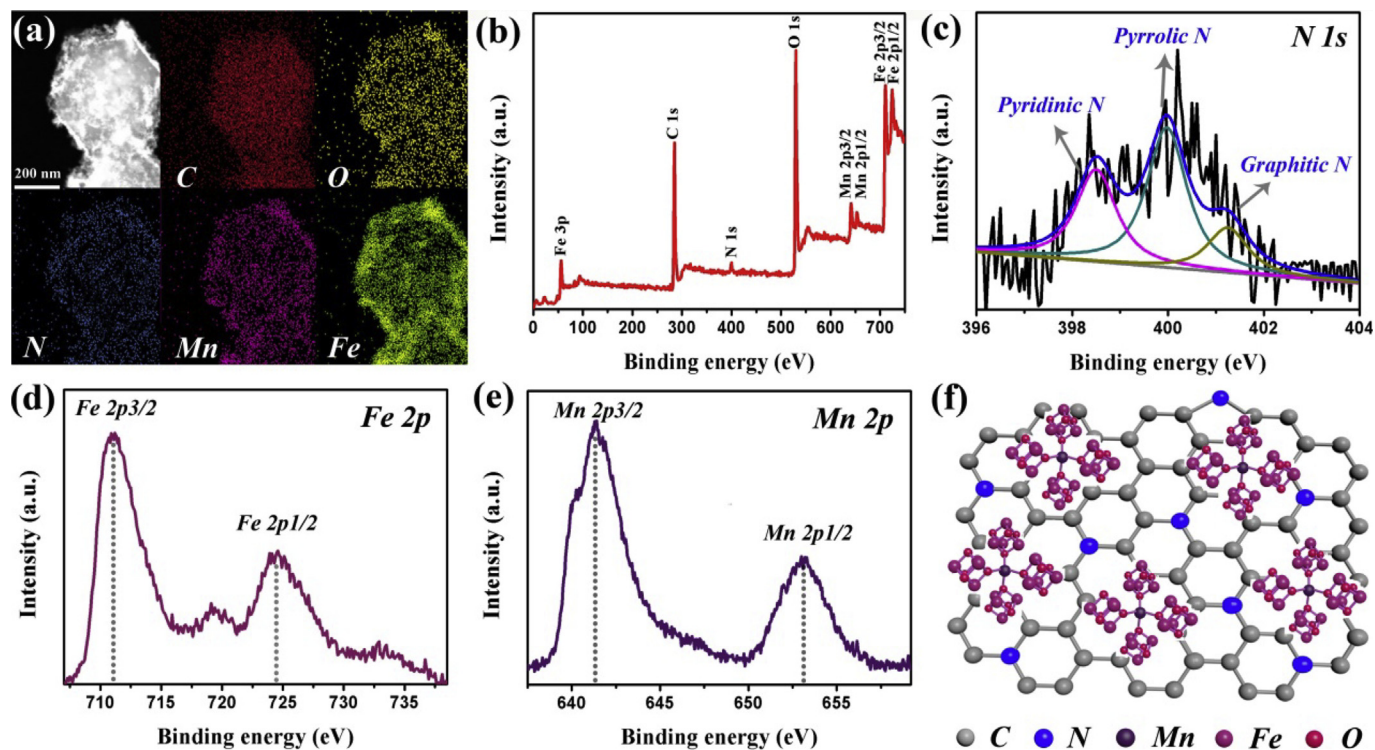


Fig. 2. (a) Elemental mapping images; (b) XPS spectrum and (c–e) the corresponding N 1s, Fe 2p and Mn 2p high-resolution spectra; (f) structural schematic diagram of 3DSC-MFQDs.

to pyridinic N, pyrrolic N and graphitic N, respectively. As an outstanding electron donor, pyrrolic N can offer high charge mobility to improve electron-transfer reactions [28]. Meanwhile, pyridinic N and pyrrolic N can easily form rich additional defects, further offering more active sites for rapid ionic kinetics of electrodes [29]. Additionally, with respect to the Fe and Mn 2p spectra in Fig. 2d and e, there are not only two characteristic peaks at 711.2 and 724.6 eV ascribed to the Fe 2p_{3/2} and Fe 2p_{1/2} of the Fe³⁺ oxidation state, but also two major peaks at 641.5 and 653.3 eV related to the Mn 2p_{3/2} and Mn 2p_{1/2} of Mn²⁺ oxidation state, which are characteristic of Fe³⁺ and Mn²⁺ of MnFe₂O₄ in MFQDs [4]. The detailed surface analysis and explanations are provided in Figs. S3 in Supplementary Materials. These results strongly indicate the successful formation of MFQDs confined in the PCPs, where a large amount of N defects are existed in the lattice of PCPs (Fig. 2f), which is beneficial for the adsorption and motion of electrolyte ions [30]. More information regarding the Raman, nitrogen absorption/desorption, pore size distribution and wettability of 3DSC-MFQDs are given in Figs. S4–7.

The electrochemical behaviors of pure MFQDs and 3DSC-MFQDs electrodes were evaluated using the three-electrode configuration in the EMIMBF₄ IL electrolyte. Fig. 3a shows the CV curves of MFQDs and 3DSC-MFQDs electrodes at a scanning rate of 50 mV/s. Prominent Faradaic redox peaks at −0.70 V (cathodic peaks) and −0.35 V (anodic peaks) are observed, suggesting that the pseudocapacitive reaction is the dominant charge-storage mechanism in MFQDs and 3DSC-MFQDs electrodes in the EMIMBF₄ IL electrolyte, which is distinguishing from the rectangular-like CV curve of PCPs with simple double-layer capacitance (dotted line in Fig. 3a and Figs. S8). The pseudocapacitive mechanism is the reversible redox reaction of MFQDs with EMIM⁺ cations in IL electrolyte, which is similar to those of previous literatures [8,15,31]. Obviously, compared with pure MFQDs electrode, the 3DSC-MFQDs electrode shows higher peak currents and larger CV area, thus revealing sufficient redox

reactions of 3DSC-MFQDs electrode in IL electrolyte [21].

The interfacial behaviors of IL ions in an electric field and specifically near the charged electrode surface are of great importance for the research of capacitive performance of electrode in IL electrolyte. The potential-driven ion accumulation probed by differential capacitance is interpreted based on Kornyshev theory, which is an indicator of IL ion density near the electrode [32]. As a consequence, the interfacial structure and differential capacitance of pure MFQDs and 3DSC-MFQDs electrodes in EMIMBF₄ IL electrolyte were studied through differential capacitance-potential (C_d -E) curves [33]. In general, electrostatic interaction leads to accumulation of counterions on the surface of charged electrode, which forms a dense Stern layer and a loose diffuse layer [34]. Differential capacitance is considered to be closely related to the ion charge density and arrangement thickness accumulated in the Stern layer [33,35,36]. Fig. 3b shows the C_d -E curves of the pure MFQDs and 3DSC-MFQDs electrodes in the EMIMBF₄ IL electrolyte, both indicating obvious characteristic of two peaks in the curves. According to the Kornyshev theory [37], for these camel-shaped curves, potential of zero charge (PZC) is the potential corresponding to the minimum capacitance between two peaks [38]. As the electrode potential increases, the BF₄[−] anions diffuse to the surface of the electrode and aggregate. Similarly, with the electrode potential negatively moving away from PZC, the EMIM⁺ cations head into the Stern layer to shield the negative charge on the electrode surface and then tightly arrange on the electrode surface. More detailed calculation method of C_d and explanation for the mechanism of camel-shaped C_d -E curves formation are provided in Supplementary Materials (Figs. S9 and 10).

However, with respect to EMIMBF₄ IL electrolyte, it is harder for the larger-sized EMIM⁺ cations to move in the diffuse layer and accumulate in the Stern layer when compared with the smaller BF₄[−] anions [34,38]. Thus, the differential capacitance of pure MFQDs electrode is lower in negative polarization than that in positive

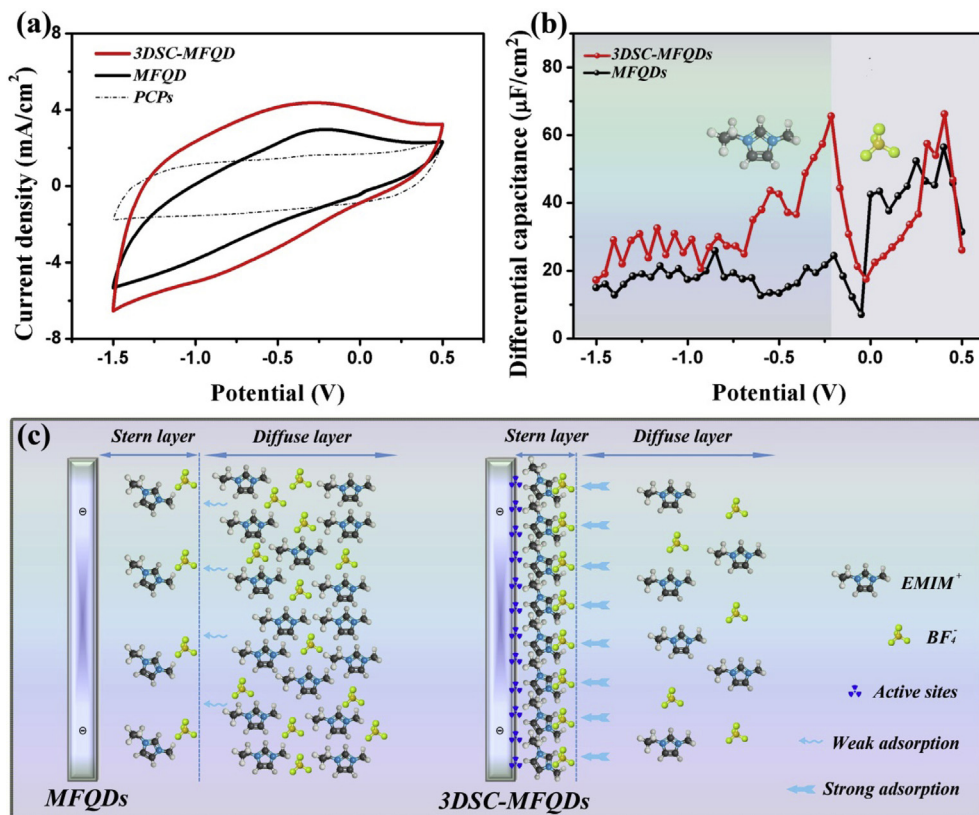


Fig. 3. (a) CV curves at a scan rate of 50 mV/s, (b) C_d -E curves and (c) schematic diagrams of Stern layer and diffuse layer structures of 3DSC-MFQDs and MFQDs electrodes in EMIMBF₄ IL electrolyte.

polarization, which represents its helpless EMIM⁺ cations accumulation in the Stern layer. Unfortunately, EMIM⁺ cation is the actual working ion which exactly participates in the pseudocapacitive reaction. Therefore, the lack of EMIM⁺ cations for pseudocapacitive reaction on the electrode surface inevitably leads to unsatisfying pseudocapacitive behaviors of pure MFQDs electrode (the left model in Fig. 3c). By contrast, the differential capacitance of the 3DSC-MFQDs electrode has almost comparable value in the negative and positive potential regions, exhibiting a significantly increased differential capacitance when the electrode is negatively charged, which is ascribed to the high accumulated ion density of EMIM⁺ cations in the Stern layer. This phenomenon could be explained by the occurrence of characteristic adsorption between 3DSC-MFQDs and EMIM⁺ cations [35]. The 3DSC-MFQDs electrode possesses plentiful active sites consisting of rich lone pair electrons, such as abundant N defects in PCPs, which tend to overlap with the electron cloud of the positively charged imidazole ring of the EMIM⁺ cation. Such sufficient overlap of electron clouds can form stronger adsorption between EMIM⁺ cations and electrode surface, ensuring that huge amounts of EMIM⁺ cations could be adsorbed as well as arrange in the Stern layer with a short range (the right model in Fig. 3c). The more EMIM⁺ cations accumulate, the more fully they partake in the redox reactions with MFQDs, which is favorable for the sufficient capacitive behaviors of 3DSC-MFQDs electrode.

EIS is a powerful technique that provides a wealth of information regarding the electrochemical kinetic characteristics of the electrode in IL electrolyte. The point intersecting the real axis in the high frequency region represents the equivalent series resistance (R_s), which reveals the total inner resistance of the electrode [39]. As shown in the inset of Fig. 4a, the R_s value of 3DSC-MFQDs

electrode is lower than that of MFQDs electrode, owing to the high electrical conductivity of PCPs in 3DSC-MFQDs electrode. Moreover, the diameter of the semicircle in high frequency region is related to the charge-transfer resistance (R_{ct}), and the straight-line part in low frequency region represents Warburg impedance (Z_w), which is described as a diffusive impedance of the IL ions within the electrode [40]. Clearly, the 3DSC-MFQDs electrode shows smaller semicircle and more vertical straight line (Fig. 4a), indicating the favorable kinetic behaviors of ionic diffusion and charge transport of 3DSC-MFQDs electrode in IL electrolyte. In order to analyze the kinetics of IL ion diffusion in depth, the diffusion coefficient of IL ions in the electrode can be estimated using impedance data at lower frequency. In this frequency range, the magnitude of the impedance varies linearly with $\omega^{-1/2}$ according to Equation (4) [41,42]:

$$Z = \frac{L}{C_L(D \cdot \omega)^{1/2}} \quad (4)$$

where D , C_L , ω and L are the diffusion coefficient, low-frequency redox capacitance, the angular frequency and electrode thickness, respectively. Value of C_L can be obtained from the plot of $-Z_{im}$ vs. $1/\omega$, which is linear with a slope of $1/C_L$, in accordance with Equation (5):

$$-Z_{im} = \frac{1}{C_L \omega} \quad (5)$$

According to Fig. 4b, the C_L values of the 3DSC-MFQDs and MFQDs electrodes were calculated to be 75.6 mF and 19.0 mF, respectively. Substituting these values as the slope in Equation (4),

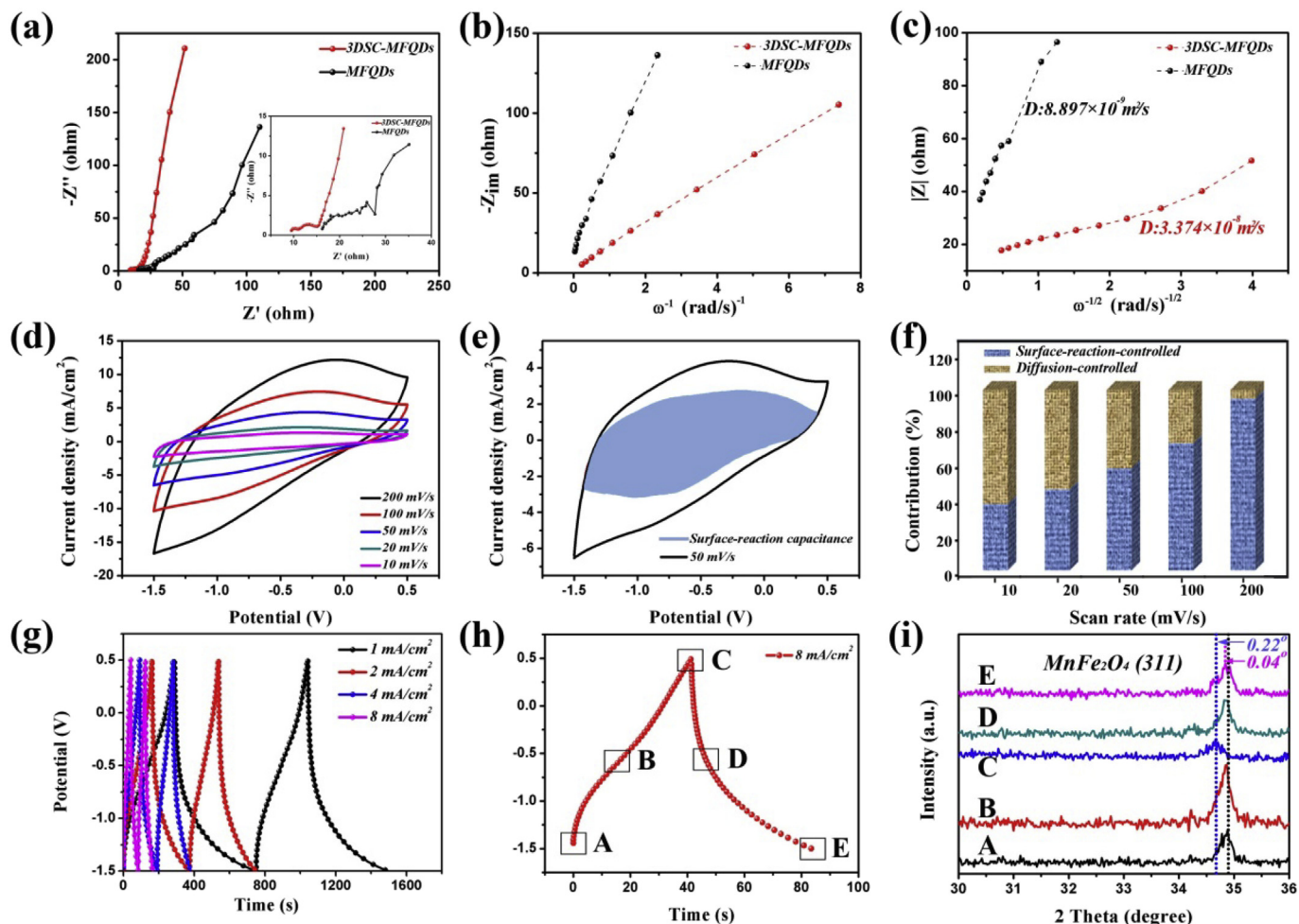


Fig. 4. (a) EIS plots; Plots of (b) $-Z''_{im}$ vs $1/\omega$ and (c) $|Z|$ vs $1/\omega$ for 3DSC-MFQDs and MFQDs electrodes. (d) CV curves; (e) voltammetric response at a sweep rate of 50 mV/s with the shaded region representing the surface-reaction capacitive contribution; (f) the capacitance contribution at different scan rates; (g) GCD curves of 3DSC-MFQDs electrode. (h) GCD curve at the current density of 8 mA/cm² and (i) electrochemical ex-situ XRD study of 3DSC-MFQDs electrode at various constant potentials in EMIMBF₄ IL electrolyte.

the diffusion coefficient of the IL ions in the 3DSC-MFQDs electrode is calculated to be 3.374×10^{-8} m²/s (Fig. 4c), much higher than that in the MFQDs electrode (8.897×10^{-9} m²/s). This result further confirms the enhanced IL ion dynamics of the 3DSC-MFQDs electrode.

Fig. 4d shows all CV plots of 3DSC-MFQDs electrode in the EMIMBF₄ IL electrolyte at various scan rates with a comparatively wide electrochemical window of -1.5 V– 0.5 V, wherein no obvious change in shape occurs at different scan rates, demonstrating its good reversibility [43,44]. To further analyze the electrochemical mechanism of 3DSC-MFQDs electrode in IL electrolyte in detail, the total charge storage was semi-quantitatively resolved by analyzing the current response of CV curves at different scan rates. As it is known, the total capacitive behavior of 3DSC-MFQDs electrode comes from surface-reaction-controlled process and diffusion-controlled process. Furthermore, the surface-reaction-controlled contribution to the total current is quantified by Equations (6) and (7) [45,46].

$$i = k_1 v + k_2 v^{1/2} \quad (6)$$

$$i/v^{1/2} = k_1 v^{1/2} + k_2 \quad (7)$$

where i , $k_1 v$ and $k_2 v^{1/2}$ represent the total current response at a

given potential, current due to surface-reaction-controlled process, and current due to diffusion-controlled process, respectively. Fig. 4e shows voltammetric response for the 3DSC-MFQDs electrode at a scan rate of 50 mV/s, wherein the surface-reaction capacitance contribution to the total current is about 56.4%, which is distinguished by the blue shaded region. As shown in Fig. 4f, the contribution of the diffusion-limited process to the total capacitance is noticeable at scan rates below 20 mV/s. However, when the scan rate exceeds 50 mV/s, the response comprises the surface capacitive effects rather than being diffusion-controlled, which is beneficial to achieving a favorable electrochemical behavior of the 3DSC-MFQDs electrode at high current densities [47]. Fig. 4g shows the GCD curves at various current densities of 3DSC-MFQDs electrode in IL electrolyte. The nonlinear curves without a plateau further show that 3DSC-MFQDs electrode mainly stores pseudo-capacitive charge in IL electrolyte [22,23]. The charge stored in the 3DSC-MFQDs electrode, or the specific capacitance, is evaluated by using GCD curves at current densities from 1 mA/cm² to 8 mA/cm². By calculating, the electrode delivers a large specific capacitance of 226 mF/cm² at a current density of 1 mA/cm² in IL electrolyte and remains a high capacitance retention ratio near 80% even at 8 mA/cm². The favorable rate capability is better than those of most previously reported monometallic oxides based electrodes in IL electrolyte (<70%) [8,15,48].

For purpose of providing insights into the electrochemical reaction of 3DSC-MFQDs and MFQDs in EMIMBF₄ IL electrolyte, ex-situ XRD analysis coupled with GCD test has been performed. Fig. 4h and Figs. S11 show the GCD curves for 3DSC-MFQDs and MFQDs at a current density of 8 mA/cm². The XRD patterns of 3DSC-MFQDs and MFQDs electrodes were recorded at various potentials (A, B, C, D and E) during the GCD cycle (Fig. 4i and Figs. S12). In the case of charging section, the XRD patterns recorded in potential region A to C clearly reveal a shift of the reflections to lower 2θ values, which manifests an increase of the lattice parameters due to the expansion of the MnFe₂O₄ host lattice caused by the intercalation of IL ions into the electrodes [49,50]. When discharged from 0.5 V back to −1.5 V, a progressive shift of the MnFe₂O₄ (311) peak to higher 2θ values occurs, evidencing a decrease of the lattice spacing, ascribed to the deintercalation of EMIM⁺ cations in the MnFe₂O₄. In Fig. 4i, the (311) peak of MnFe₂O₄ for 3DSC-MFQDs electrode initially observed at 33.14° at −1.5 V shifts to 32.92° when charged to 0.5 V. Afterwards, the (311) peak shifts to 33.1° when discharged to −1.5 V, which is virtually the same as its original 2θ position, demonstrating a relatively reversible transition within 3DSC-MFQDs electrode during the charge-discharge cycle (Fig. 4i) [44,51]. By contrast, the (311) peak of MnFe₂O₄ for MFQDs exhibits a large irreversible left shift (0.13°) during the charge and discharge cycle (Figs. S12). It comes to the conclusion that the structural stability of 3DSC-MFQDs electrode has been effectively enhanced during the electrochemical reactions, resulting in high rate capability especially at high current densities. Moreover, the cycling performance has also been promoted contributing from durable structural stability (Figs. S13 and 14). More information about electrochemical performance of 3DSC-MFQDs, MFQDs and PCPs electrodes is shown in Figs. S15 and 16.

The superior electrochemical behaviors of the 3DSC-MFQDs electrode in IL electrolyte are owing to the 3D space-confinement supported by PCPs, in which porous and conductive PCPs act as ion-accessible channel and charge-transfer pathway (Fig. 5). The merits of 3D space-confinement supported by PCPs are: (1) Abundant active sites consisting of numerous lone pair electrons, especially nitrogen defects in PCPs, are favorable for the adsorption of IL ions. Meanwhile, rich porous structure of PCPs provides accessible channels for IL ions (blue arrows). Both of them ensure IL ions involved in the sufficient redox reaction with MFQDs, leading to the high specific capacitance of 3DSC-MFQDs electrode. (2) Not

only are the MFQDs steadily confined in PCPs to form a robust 3D architecture, which effectively inhibits the aggregation and dissolution of MFQDs during the cycling processes, but rich mesoporous compartments from PCPs also provide adequate buffer space for accommodating the volume change of MFQDs during repeated charge-discharge processes, rendering high cyclic stability of the 3DSC-MFQDs electrode. (3) Highly conductive PCPs act as the “highway” to facilitate the efficient charge transport (red arrows), which is favorable for the high rate capability of 3DSC-MFQDs in IL electrolyte. The ionic diffusion and charge transport of the 3DSC-MFQDs electrode in IL electrolyte are provided in Supplementary Movie 1.

Supplementary video related to this article can be found at <https://doi.org/10.1016/j.electacta.2019.135386>

For practical application, a kind of flexible SCs (FSCs) based on ionogel electrolyte were fabricated, in which 3DSC-MFQDs and commercial activated carbon (AC, Figs. S17–19) pressed on flexible graphite paper were selected as negative and positive electrodes, respectively (Fig. 6a). The FSCs have robust flexibility and can be operated in a wide working voltage window between 0 and 3.5 V (inset in Fig. 6a and Figs. S20). The CV curves display a quasi-rectangular and mirror shape without obvious distortion corroborating the high reversibility and rate capability (Fig. 6b). Calculated from the GCD curves (Fig. 6c) of the as-fabricated devices at different current densities, the FSCs deliver a maximum energy density up to 67.6 W h/kg (68.0 μW h/cm²) at a power density of 1.75 kW/kg (1470 μW/cm²) (Fig. 6d and Figs. S21). Such an energy storage performance is comparable to that of Li-ion/NiMH/Lead-acid batteries, and superior to that of most previously reported Mn or Fe based aqueous FSCs (most <25 W h/kg, see Table S1 for comparison) [52–57] and commercial SCs (~5 W h/kg) [58]. Meanwhile, the device also exhibits durable cyclic stability with high capacitance retention (>80%) at 8 A/g after 10000 cycles under both straight and bending conditions (Fig. 6e). After full charging, the FSCs can easily light up the illuminations for a couple of minutes, which confirms their promising potential as energy storage devices (Fig. 6f and Supplementary Movie 2). As a demonstration of the potential applications in flexible electronic devices, two FSCs are connected in parallel arrangement and effectively power wearable electronics including a smart bracelet, even in complex motion states. EIS result about the as-assembled FSCs is shown in Figs. S22. Therefore, this work has put forward an intriguing 3D

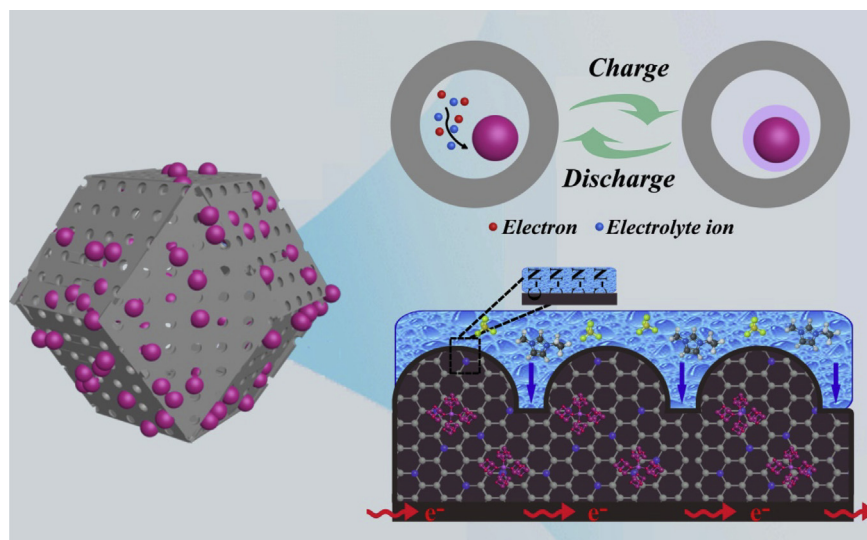


Fig. 5. Schematic diagram of ionic diffusion and charge transport in 3DSC-MFQDs electrode during the process of charge and discharge.

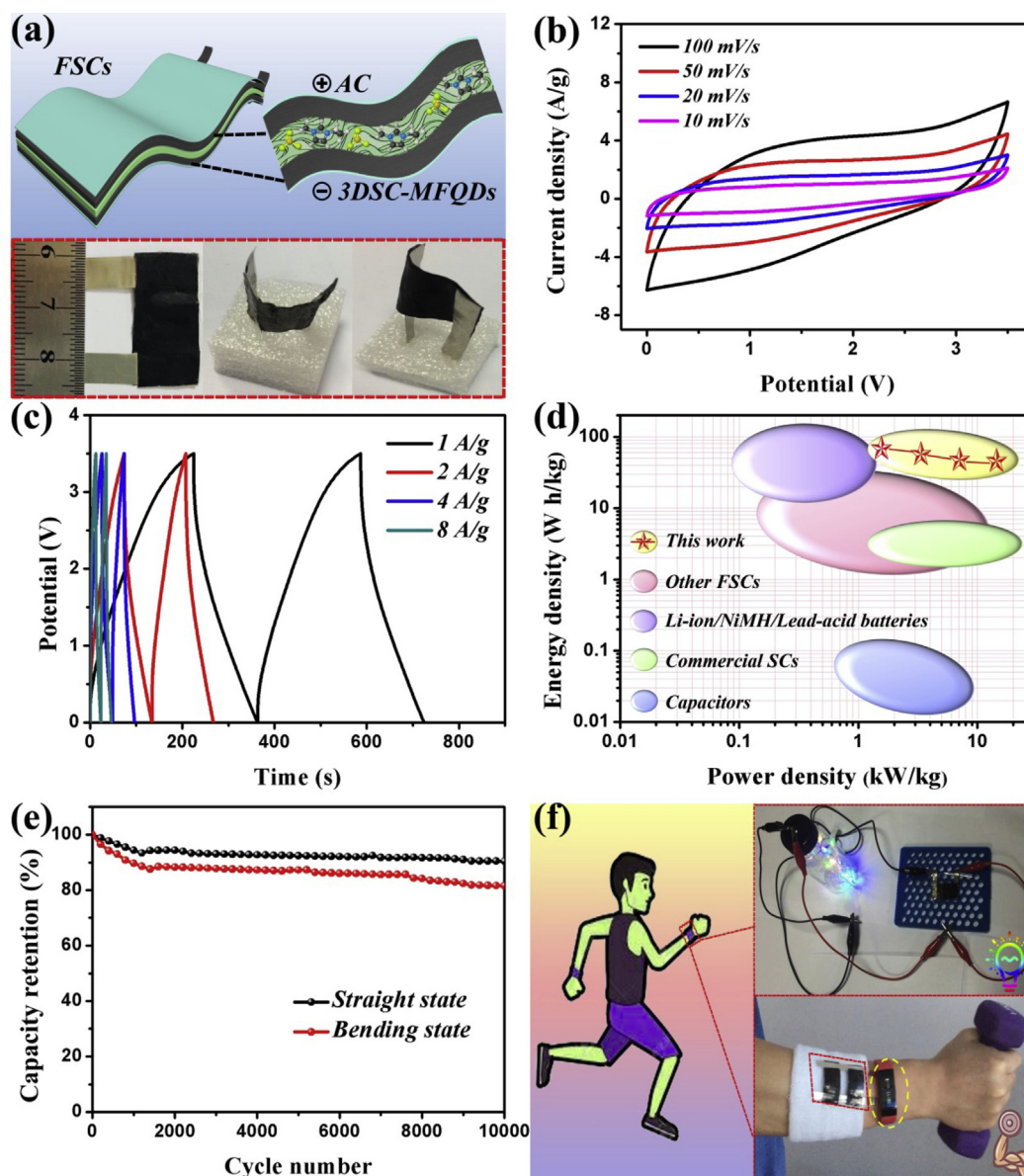


Fig. 6. (a) Fabrication procedure of 3DSC-MFQDs//AC FSCs. The inset in a shows digital photos of 3DSC-MFQDs//AC FSCs under straight and bending states. (b) CV curves at various scan rates; (c) GCD curves at different current densities of 1–8 A/g; (d) Ragone plot; (e) Cycle performance of 3DSC-MFQDs//AC FSCs at the current density of 8 A/g after 10000 cycles. (f) The power supply of 3DSC-MFQDs//AC FSCs to colorful illuminations and wearable electronics in complex motion states after fully charged.

space-confined Mn–Fe bimetallic oxides electrode for FSCs with appealing potential to high performance energy storage devices.

4. Conclusions

In summary, we have successfully developed a unique 3DSC-MFQDs electrode and conducted an in-depth interfacial electrochemical investigation of 3DSC-MFQDs in IL electrolyte. Remarkably, PCPs not only act as ion-accessible channel and charge-transfer pathway, but also provide a 3D nanospace for confining MFQDs to ensure the structural stability of the electrode. Importantly, profiting from the characteristic adsorption of electrode and EMIM⁺ cations of IL electrolyte which may appear in other imidazole-based IL electrolytes, superior interfacial electrochemical behaviors and highly reversible pseudocapacitive storage could occur in 3DSC-MFQDs electrode, demonstrated by the

potential-driven ion accumulation probe. On this basis, the as-assembled FSCs deliver an attractive energy storage capability (energy density up to 67.6 W h/kg at a power density of 1.75 kW/kg) and durable cycling life under both straight and bending conditions, which represent their promising electrochemical reliability under hard mechanical conditions. More importantly, the as-fabricated devices can effectively supply power to wearable electronics even in complex motion states. Hence, this work can throw new light on advanced MnFe₂O₄ based materials and their interfacial electrochemical investigation for flexible energy storage devices to meet diverse demands of energy storage systems in portable and wearable electronics.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.electacta.2019.135386>.

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